Project Overview

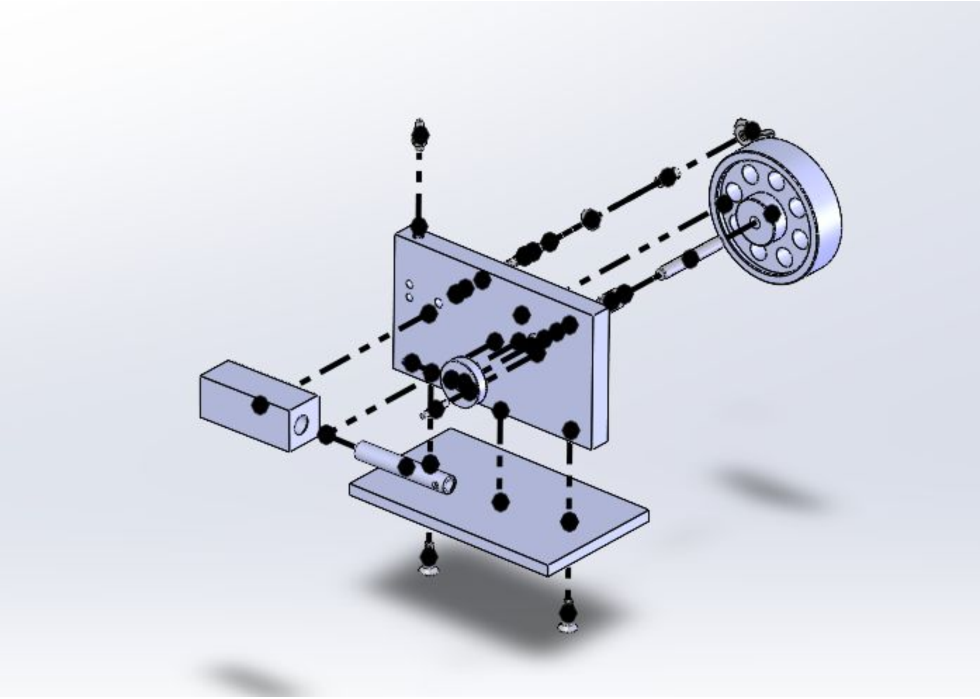
Goal:
Build a Wobbler Engine that would run reliably at competition

Criteria:
- Operation at 20 psi for over 5 minutes
- Smooth motion
- Leakage controlled

Responsibilities:
- CAD
- Machining parts to certain tolerances
- Assembly alignment and fit
- Test and tuning

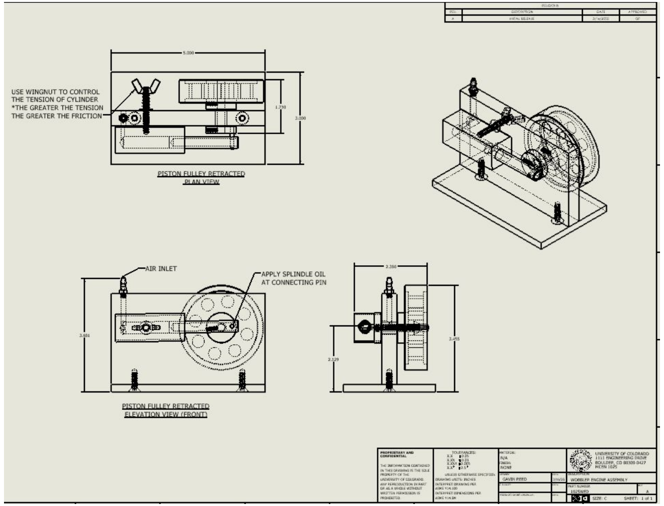
Systems and Interfaces

- Designed 15 parts on SolidWorks and got screws off McMaster
- Made drawings for every single part including tolerances
- Made BOM and full assembly drawings with smooth motion to emulate the real design



Build and Integration

- Machined most of the parts using mill and lathe with using the drawings as reference to ensure accuracy
- Fully assembled the parts to make the engine
- Difficulties with some interfaces in which I isolated the problem and slightly changed the dimensions to ensure full function



Testing and Tuning

- Hours of testing to ensure best performance
- Hand forced testing
- Used water to visibly check for any unwanted leaks
- Adjusted wing nut to ensure no leakage from increasing pressure but also limited friction

Results and Lessons Learned

Results:

- Easily got 20 psi for over 5 minutes
- Did well at competition lasting until top 5

Lessons Learned:

- Documentation is so important
- Very precise with detail during design
- Ensuring integration between parts is vidal